

Google Play Store Assignment Report

1. Challenges Faced During Preprocessing

During this assignment, I faced several challenges while preprocessing the Google Play Store dataset.

- The dataset contained duplicate records, which were removed before further analysis.
- Some numerical columns such as Installs, Price, Size, and Reviews were stored as text and required cleaning before converting them into numeric format.
- The dataset had missing values in both numerical and categorical columns, so different imputation methods were applied.
- Outliers were present in columns like Rating, Reviews, Installs, and Price, which were detected using the IQR and Z-score methods.
- Categorical columns had to be encoded before they could be used in machine learning.
- New features had to be created without affecting the original dataset.

2. Important Features Discovered

- Reviews
- Installs
- Rating
- Price
- Reviews_Per_Install
- Paid_App_Flag
- App_Age
- Popularity_Score
- Revenue_Potential
- Category_Average_Rating

These features can help in understanding app popularity and performance.

3. Key Business Insights

- Free apps have more installs compared to paid apps.
- Apps with more reviews generally have higher popularity.
- Rating is related to user engagement and reviews.
- Revenue potential depends on both the app price and the number of installs.
- Category average rating helps compare the overall performance of different app categories.

4. Final Conclusion

The Google Play Store dataset contained missing values, duplicate records, inconsistent formats, and outliers.

After performing data preprocessing, the dataset became clean and suitable for machine learning.

Feature engineering added useful columns such as Reviews_Per_Install, Paid_App_Flag, App_Age, Popularity_Score, Revenue_Potential, and Category_Average_Rating, which can improve future analysis and prediction models.